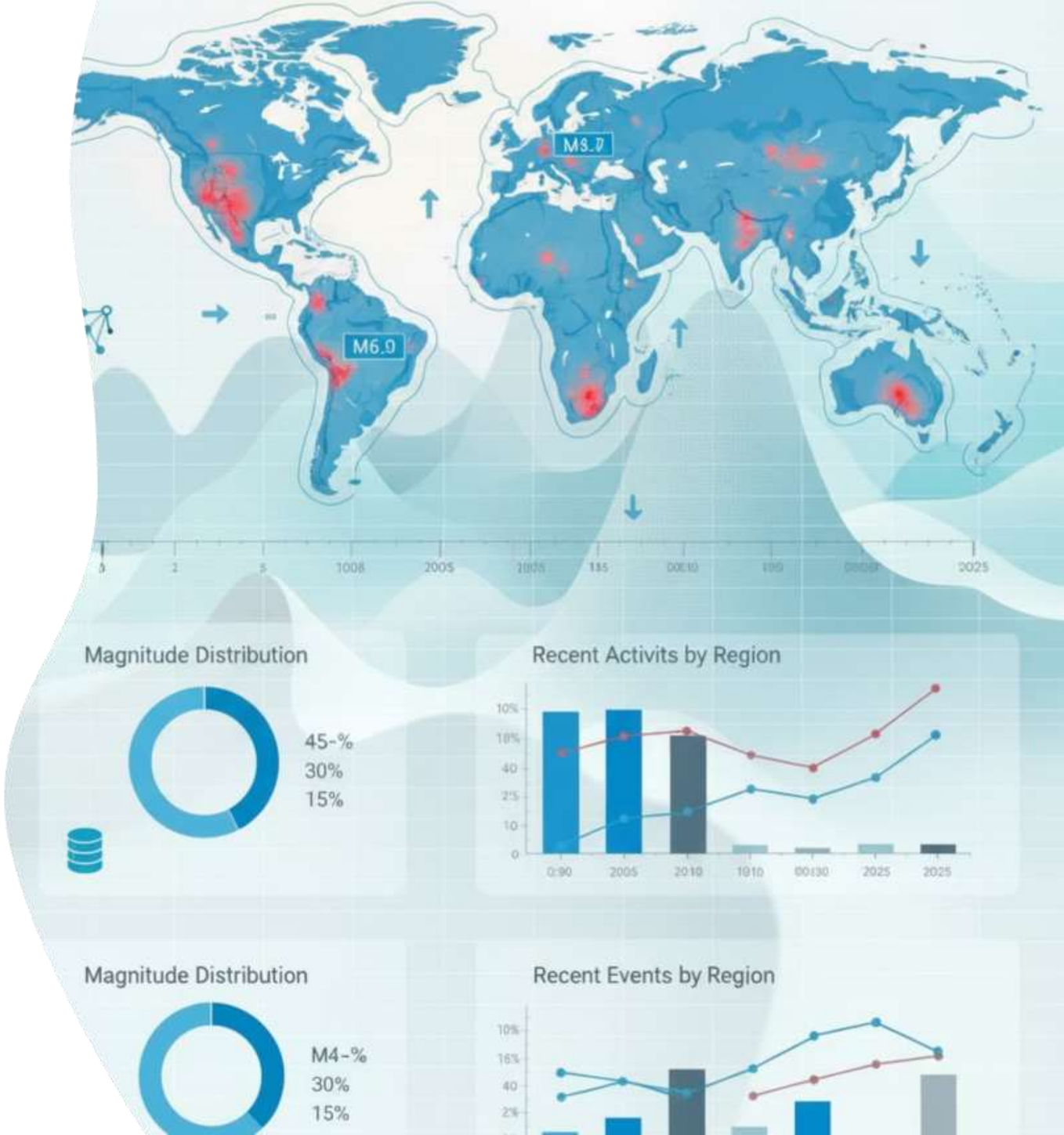
Global Seismic Trends: Data-Driven Earthquake Insights



Understanding the project

Domain Context

Seismology and disaster analytics play a critical role in understanding earthquake behaviour patterns and assessing seismic risk across regions.

Project Focus

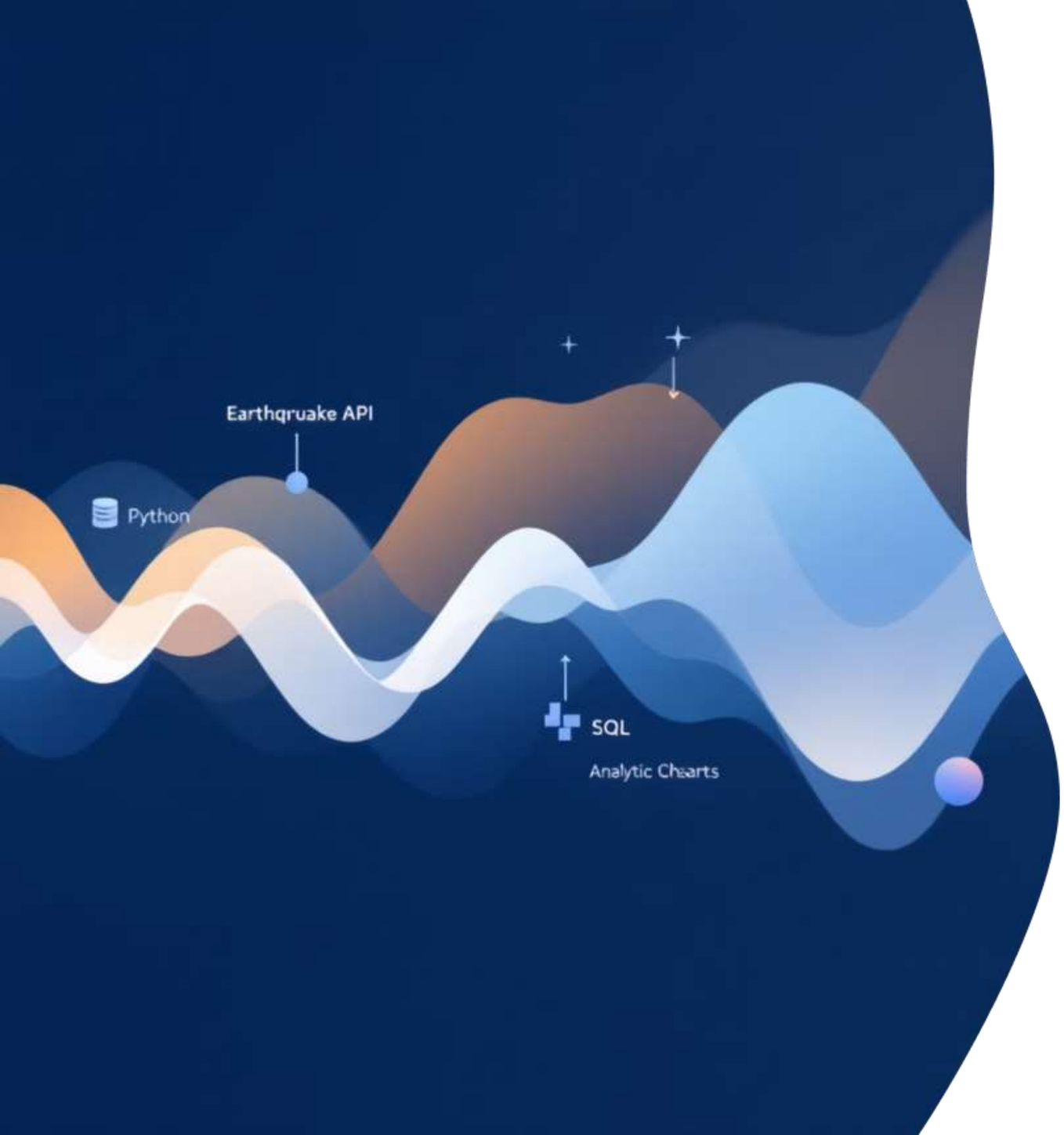
Comprehensive analysis of global earthquake data through systematic API extraction, database management, and SQL-driven analytics to uncover meaningful patterns.

Problem Statement

Analyse global earthquake data to identify seismic patterns, temporal trends, high-risk geographical zones, and deliver actionable insights for disaster preparedness.

Business Impact

Support governments, insurers, and researchers in earthquake risk assessment, disaster planning, and data-driven decisions for urban safety and infrastructure resilience.



Data Collection & Preparation Pipeline

API Retrieval

Retrieved global earthquake records from public API using Python for selected time period with comprehensive parameters.

Data Cleaning

Cleaned and preprocessed data by standardizing timestamps, managing missing values, and ensuring data quality across all fields.

Feature Engineering

Created new analytical columns to support time-based and impact-based analysis, enhancing analytical potential.

Data Readiness

Organised the dataset to ensure consistency, completeness, and analytical readiness for advanced seismic pattern analysis.

Analysis Framework & Insights

Database Architecture

Stored the processed earthquake dataset in a MySQL database with optimised schema design for efficient querying and analysis.

Analytical Approach

Applied SQL queries to analyse seismic trends and regional activity patterns. Examined earthquake behaviour across multiple dimensions including time, depth, and magnitude to uncover hidden patterns.

Insight Generation

Converted complex analytical results into clear, interpretable insights that support evidence-based decision-making for disaster preparedness.

Core Technical Competencies

Python & Data Engineering

- API data retrieval and integration
- Pandas for preprocessing and cleaning

SQL & Database Management

- Schema design and optimisation
- Complex analytical querying

Visualisation & Dashboards

- Streamlit interactive dashboards
- Dynamic filtering capabilities

Analytical Excellence

- Exploratory Data Analysis (EDA)
- Logical thinking and reasoning
- Deriving meaningful insights from raw data

Impact & Outcomes

1

End-to-End Pipeline

Successfully implemented a complete analytics pipeline from API data extraction through to actionable insights, demonstrating technical proficiency across the data lifecycle.

2

Risk Assessment Framework

Applied data-driven analysis specifically designed for seismic risk assessment and disaster planning, enabling stakeholders to make informed decisions.

3

Stakeholder Value

Delivered meaningful insights that support governments, insurers, and researchers in protecting communities, optimising infrastructure resilience, and enhancing urban safety measures.

> A comprehensive demonstration of data engineering, analytical thinking, and domain expertise in seismic risk analytics.